A Multilayered Adaptive Recurrent Incremental Network Model for Heterogeneity-Aware Prediction of Derived Remote Sensing Image Time Series

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***Abstract*— *Catastrophic forgetting* of previously acquired knowledge is a major setback suffered by the neural networks (NNs) when these are trained on tasks in sequential fashion. The NN variants, including the deep network models, as commonly used in remote sensing data prediction are also not free from this limitation. The issue becomes more prominent when the prediction is performed over data collected from spatial zones with a large degree of subregional variations or heterogeneity. In order to tackle this problem, in this article, we propose a multilayered adaptive recurrent incremental network (MARINE) model, offering a spatial heterogeneity-aware self-adaptation scheme that resolves the catastrophic forgetting issue in an autonomous manner. Typically, the proposed MARINE is equipped with an intrinsic mechanism of clustering the spatial subregions as per their heterogeneity levels and auto-constructing the recurrent network layers in an ensemble fashion so that the knowledge acquired about one group of heterogeneity level does not overwrite that acquired about other groups. With respect to spatio-temporal prediction of normalized difference vegetation index (NDVI) time series, as derived from MODIS Terra satellite remote sensing imagery, we demonstrate that our proposed MARINE achieves competitive results when compared with the state-of-the-arts. More significantly, in the presence of higher degree of spatial heterogeneity, MARINE outperforms others by better *avoiding the catastrophic forgetting issue*.**

***Index Terms*— Autonomous learning, ensemble model, remote sensing, spatial heterogeneity, spatio-temporal prediction.**

I. Introduction

APPLICATION of deep neural networks (NNs) for spatio-temporal prediction of remote sensing data has gained immense popularity in last few decades [1]. However, apart from the issues like lack of labeled data, inappropriate network configuration, etc., the performances of the existing models are often found to deteriorate because of the presence of varying degrees of *heterogeneity* or *subregional variances* throughout the study region [2]. Typically, the spatial heterogeneity can be defined as "uneven distribution of a trait, event, or relationship across a region" [3]. This can severely affect any NN-based prediction model. As illustrated in Fig. 1, the spatial relationships among the pixels within each subregion in the spatial Zone-A are more or less similar, and, whatever be the pixel values, these show a high degree of positive correlation between nearby as well as distant pixels in every case. Contrarily, in the case of spatial Zone-B, the spatial relationships among the pixels within each subregion vary notably. These significantly different relationships at the various subregions in Zone-B prominently indicates the presence of spatial heterogeneity, whereas the subregions in spatial Zone-A are quite homogeneous or have comparatively low heterogeneity. A popular approach to handle such spatial heterogeneity is to use separate (local) models for each of

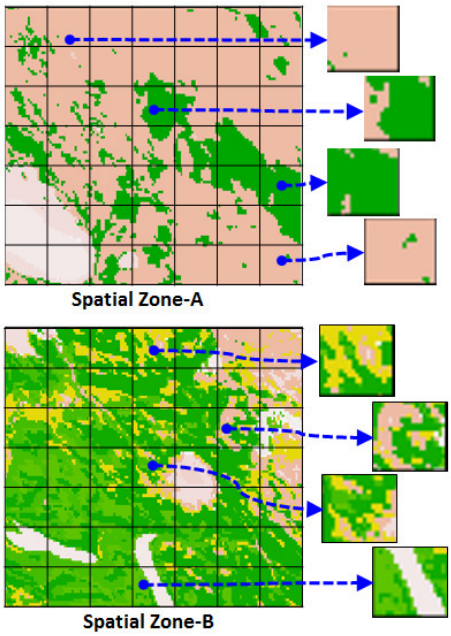


Fig. 1. Example of two spatial regions/zones (Zone-A and Zone-B) having different levels of spatial heterogeneity throughout the containing subregions.

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these subregions [4]. However, this can make the overall approach challenging in terms of increasing computational burden. Otherwise, if one single model is employed for all the subregions, the relationships learned by the model from one subregion may be overwritten by the subsequent learning of significantly different relationship from another subregion. This phenomenon of overwriting acquired knowledge during sequential learning is technically termed as ***catastrophic forgetting*** [5]. Our goal is to come up with a single model that can handle this catastrophic forgetting issue while tackling heterogeneity in spatial time series data.

In particular, to combat the above-discussed issue, we propose a variant of multilayered adaptive recurrent NN (RNN) model which is able to incrementally learn the varying degrees of subregional relationships by employing separate network layers in ensemble fashion. The idea of using ensemble architecture is motivated by its potentiality to not only avoid catastrophic forgetting [5], [6] but also allow quick learning from limited data samples, unlike the conventional deep hierarchical architectures. It must be mentioned here that the prediction task as considered in this article is primarily *spatio-temporal regression* of single-sensor data derived from satellite remote sensing image time series.

A. Related Works

1) Deep Learning-Based Prediction Models: Exploiting the power of deep NNs for *spatio-temporal prediction* of remote sensing data has attracted significant attention in recent years. Compared to the various conventional techniques, such as spatio-temporally weighted regression [7], space-time Kriging [8], and Kalman filtering [9], the deep learning-based approaches [10]–[13] have been found to be more advantageous with respect to two major viewpoints. First, using their deep/stacking architectures, the deep network models are able to hierarchically extract features without any manual feature engineering [10], [11], [14]. Second, the deep networks with recurrent architecture have additional benefit of effectively remembering the temporal sequences during time series prediction [12], [13]. For example, the RNN model with long short-term memory (LSTM) architecture, as used in the work of Zhang *et al.* [12], has been found to perform well in predicting sea surface temperature using high-resolution satellite remote sensing images. The recently proposed two stacked LSTM along with added layer of multilayer perceptron (MLP) [15] has shown more promising performance in this respect. As per the research of Yang *et al.* [11], the LSTM model with combined convolutional layer is also effective for this purpose. Incidentally, in the present context of spatial image time series prediction, the combination of LSTM with the convolutional layer is found to be more suitable than simple convolutional NN (CNN) models [16], since the former can exploit the power of LSTM to capture the temporal change in the dataset. Similar inference can also be drawn from the recent works by Sun *et al.* [17] and Nevavuori *et al.* [18], who proposed two other variants of CNN-LSTM network for spatio-temporal prediction of crop yield from remote sensing data. Apart from these CNN and LSTM-based approaches, the deep stacking network models, such as Deep-STEP [10], Deep-STEP_FE [14], etc., have also been found to perform well in predicting derived remote sensing image for future and intermediate timestamps, respectively. However, none of these models is designed to handle *catastrophic forgetting issue* arising due to *spatial heterogeneity condition*, and hence, when applied on subregional basis, the performance of these models deteriorates considerably. Besides, the iterative learning requirement and the blind architecture selection are two other major issues encountered by these deep models, and consequently, these often fail to deliver the desired prediction accuracy on time. Though the recently proposed SARDINE model [1], addresses the iterative learning and rigid architecture challenges, the self-evolution scheme of SARDINE is also not appropriate for dealing with *high* spatial/spatio-temporal heterogeneity.

2) Prediction Models Handling Spatial Heterogeneity Issue: Practically, the spatio-temporal modeling under spatial heterogeneity condition is quite a little-explored research area so far [2]. Though Georganos *et al.* [2] and Yang *et al.* [11] have recently attempted to handle spatial heterogeneity by decomposing global process into several local submodels, this does not completely address the issue. As illustrated in our introductory section, the use of individual local model can make the approach computationally expensive, whereas the sequential processing of local submodels may lead to serious catastrophic forgetting problem [5], especially for any NN model having either shallow or deep architecture. To the best of authors' knowledge, ours is the first work on multilayered NN that explicitly deals with this challenge under the background of remote sensing image time series prediction. Note that, the spatial ensemble learning (SEL) [19] and the spatial variability-aware deep NNs (SVANNs) [4] are two recent models that are quite relevant to this topic. However, our proposed model is prominently distinguishable from these works with respect to ***three major perspectives***: 1) SEL and SVANN are defined for classification purpose; not for the present context of regression task; 2) unlike our model, these are fundamentally developed for dealing with spatial data, and hence, need further enhancement to deal with spatio-temporal data, finally; and 3) SEL and SVANN require separate deep learning model (classifier) for each subregion/partition which leads to high model complexity. Contrarily, in our case, a single learning model maintains separate layers to handle different groups of subregions having similar degree of heterogeneity, and thus, the model complexity is reduced to a great extent.

B. Contributions

Motivated by the research gaps in the state-of-the-arts, in this article, we propose a heterogeneity-aware spatio-temporal prediction framework, which is grounded on autonomous recurrent learning principle [1], [6], [20]. In our proposed prediction framework, the awareness to spatial heterogeneity is achieved by means of using a multilayered adaptive recurrent incremental network, hereafter termed as MARINE. The incremental approach of learning through

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multilayered ensemble architecture helps MARINE to tackle catastrophic forgetting issue, whereas, the self-guided network evolution as well as autonomous threshold adjustment power of MARINE aids the model to attain online prediction while diminishing the requirement of manual interaction. It may be noted that, though MARINE is influenced by the concept of self-adaptive/autonomous RNN from our earlier works [1], [6], it has its own novelties as well, in following respects.

1) In contrast to [1], MARINE requires no explicit process of generating spatial feature on per-pixel basis, and hence, it is straightaway applicable on the derived remote sensing image streams.
2) Unlike [6], the layer growth principle in MARINE is defined exclusively for the purpose of capturing the diversity in spatial heterogeneity level. Moreover, the threshold adjustment scheme in MARINE is purely autonomous.
3) Most significantly, MARINE is equipped with a unique mechanism of handling spatial heterogeneity which is not very well-addressed in earlier works.

Thus, the major contributions of this work are as follows.

1) Introducing a *spatial-heterogeneity-aware framework* for spatio-temporal prediction of remotely sensed geo-spatial image time series.
2) Proposing a variant of multi-layered adaptive recurrent incremental network (MARINE), which has intrinsic ability to autonomously learn spatial/spatio-temporal relationships while better avoiding the catastrophic forgetting due to spatial heterogeneity.
3) Providing a principled approach of *self-constructing* as well as *characterizing* the network-layers in the ensemble architecture of self-adaptive RNN.
4) Defining a method of *auto-adjusting the threshold value* to ease the self-decision-making process regarding the growth of network-layer and to further strengthen the autonomous learning feature of the proposed model.
5) Validating the proposed MARINE-based prediction framework using real-world datasets derived from MODIS Terra satellite remote sensing images.

The results of experimentation, considering spatial regions with varying degrees of spatial heterogeneity, demonstrate the effectiveness as well as superiority of MARINE over the state-of-the-art deep network models.

The rest of the article is organized as follows. The problem formulation is presented in Section II. The proposed heterogeneity-aware prediction framework along with the detailed working principle of MARINE is discussed in Section III. The experimental evaluation of our proposed model is presented in Section IV. Finally, the concluding remarks are made in Section V.

II. Problem Formulation

This section formulates the overall prediction problem from both genetic and subregional perspectives.

Given the spatial image time series $I = \{I_1, I_2, \ldots, I_t\}$ where each $I_i \in \mathbb{R}^N$ is over any variable (e.g., soil moisture, land surface temperature (LST), and vegetation index) as derived from the satellite remote sensing imagery over the time stamps $T_1, T_2, \ldots, T_t$, the goal/task is to predict I_{t+1} such that

$$\hat{I}_{t+1} = \arg\max_{I_{t+1}} \Pr(I_{t+1} | I_1, I_2, \ldots, I_t). \quad (1)$$

Here, $\hat{I}_{t+1}$ is the predicted image for the time stamp T_{t+1}. Thus, the generic objective function becomes as follows:

$$\min_{\hat{I}_{t+1} \in \mathbb{R}^N, I_{t+1} \in \mathbb{R}^N} \mathcal{L}(\hat{I}_{t+1}, I_{t+1}) \quad (2)$$

where $\mathcal{L}$ represents the loss function, and N denotes the total number of pixels in a single image I_i at time stamp T_i. Since our proposed model is defined to work on subregional basis, the objective function above can be approximated in a manner as formalized below

$$\min_{\hat{S}(s,t+1) \in \mathbb{R}^n, S(s,t+1) \in \mathbb{R}^n} \sum_{i=1}^{R} \mathcal{L}(\hat{S}(s_i, t+1), S(s_i, t+1)) \quad (3)$$

where n is the pixel count in a subregion, $R = N/n$ is the total count of subregions that compose a single image in I, $S(s_i, t+1)$ indicates the ith subregional data in the spatial image at time T_{t+1}, and $\hat{S}(s_i, t+1)$ is the corresponding predicted data which can be expressed as follows:

$$\hat{S}(s_i, t+1) = \arg\max_{S(s_i, t+1)} \Pr(S(s_i, t+1) | S(s_i, 1), \ldots, S(s_i, t)). \quad (4)$$

The concept of subregion is pictorially illustrated in Fig. 2.

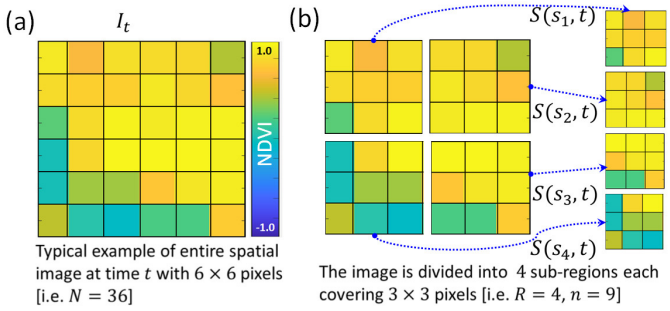


Fig. 2. Illustration for (a) spatial region and (b) its sub-regional divisions.

III. Proposed Prediction Framework Based on MARINE

This section illustrates the proposed heterogeneity-aware prediction framework which attempts to solve the above-discussed problem by means of a multi-layered adaptive recurrent incremental network, termed as MARINE.

A. Model Overview

The flow of processes within the proposed framework is depicted in Fig. 3. As shown in the figure, the proposed prediction model works on subregional basis, and accordingly, assumes each image of the study zone to be subdivided into several subregions of equal spatial extents. The model starts with measuring the spatial heterogeneity of each subregion, which is subsequently used to make layer growing decision at the time of learning/prediction through a self-adaptive RNN model, termed as MARINE. Each of these layers in MARINE

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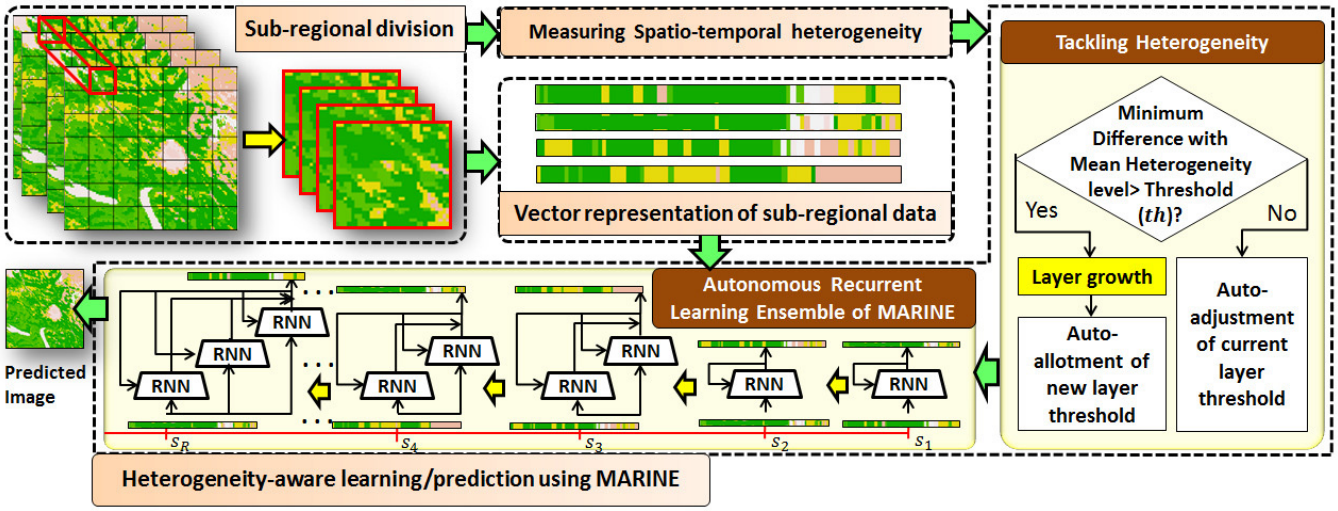


Fig. 3. Proposed MARINE-based heterogeneity-aware spatio-temporal prediction process.

helps the model in separately learning heterogeneous spatial relationships from various subregions, and thereby, judiciously handles the possible catastrophic forgetting issue. In order to facilitate the RNN-based analysis, the model converts the data from each subregion into respective vector representation, in a preprocessing step. Overall, our proposed prediction model is composed of four basic modules corresponding to *sub-regional division of data*, *measuring spatial/spatio-temporal heterogeneity*, *vector representation of the subregional data*, and finally, *heterogeneity-aware learning and prediction using MARINE*. The technical details for each of these modules are presented in Sections III-B–III-E.

B. Subregional Division of Data

The objective in this step is to divide each image $I_j \in \mathbb{R}^N$ of the study area at time stamp T_j into R number of subregional images s_i, $i = 1, \cdots R$, each containing n number of pixels. Therefore, $R = N/n$. Formally, to accommodate both spatial and temporal information, we denote such subregional data as $S(s_i, j)$, and accordingly, $I_j = \cup_{i=1}^{R} S(s_i, j)$. The idea can be visualized from the top-left block in Fig. 3 and also from Fig. 2. The subregional division of the data (image time series) is primarily inspired by the fact that the pixels situating at far apart distances are likely to be less related to each other [21], and hence, learning from the entire region may provide simply a gross view of the region to the model [2]. Accordingly, learning from subregional data offers more facility to better capture the spatial relationships from the data. To be noted, this facility is by default entertained in CNN-based models because of their intrinsic filtering mechanism. However, as highlighted through a motivating example in the introductory section, learning from subregional data may lead to problem of catastrophic forgetting due to varying degree of spatial heterogeneity throughout the subregions. Our proposed framework is designed to address this issue by exploiting the power of autonomous machine learning as explained in Sections III-E1–III-E3.

C. Measuring Spatial/Spatio-Temporal Heterogeneity

There are various ways of measuring spatial heterogeneity in terms of correlation, entropy, standard deviation, etc., [22]. In our model, the spatial heterogeneity within each subregion is measured in terms of "*range*," as obtained through variogram analysis which provides a quantitative as well as descriptive statistic of the spatial continuity of the dataset [3], [23]. Typically, a spherical type (default) variogram model over a subregion s_i can be presented as follows:

$$\gamma_i(\mathcal{D}) = \begin{cases} \gamma_i(0) + \chi_i \cdot \left[1.5\left(\frac{\mathcal{D}}{a_i}\right) - 0.5\left(\frac{\mathcal{D}}{a_i}\right)^3\right], & \mathcal{D} \le a_i \\ \gamma_i(0) + \chi_i, & \mathcal{D} > a_i \end{cases} \quad (5)$$

where $\mathcal{D}$ indicates the spatial lag distance vector; $\gamma_i(0)$ represents the *nugget* or the discontinuity at the origin; χ_i represents the *sill* or the variance at which the variogram stops changing; and a_i denotes the *"range"* or the spatial lag distance at which the variogram reaches the plateau and stops changing. This *"range" value* can be considered as a measure for *degree/level of spatial heterogeneity* [3]. The lower is the range, the more heterogeneous is the subregion.

In order to take into account the temporal variation of the subregional data while maintaining the computational time for the above-mentioned variogram analysis, we first consolidate the subregional data $S(s_i, j)$ over all the observed time stamps T_j, $(j = 1, \ldots, t)$ using weighted average as follows: $cons_i = \sum_{j=1}^{t} TW_j \times S(s_i, j)$. Here, $TW_j = ((id_j)/(\sum_{k=1}^{t} id_k))$, where $id_j = (1/t - j + 1)$ indicates the inverse temporal distance of the time stamp T_j from the prediction time stamp $T_{pred} = T_{t+1}$. This helps to take into account the temporal auto-correlation of the data [24]. Subsequently, the consolidated data $cons_i$ is used for estimating the subregional heterogeneity as defined in (5).

D. Representing Subregional Data in Vector Form

This module aims at representing each subregional data $S(s_i, j)$ at time stamp T_j $(j = 1, \ldots, t)$, into vector of

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dimension $(1 \times n)$ (refer Fig. 3). Formation of such vectors helps the subregional data to be fed at a time as input to the RNN-based autonomous learning in the succeeding module.

E. Heterogeneity-Aware Learning/Prediction Using MARINE

The key objective of this module is to model the temporal evolution of spatial features within each subregion by using a MARINE. The MARINE works in collaboration with a heterogeneity tackling unit to offer a unique scheme of heterogeneity-aware learning/prediction of spatial image time series, which eventually helps mitigating the issue of catastrophic forgetting. Typically, whenever the heterogeneity tackling unit identifies the heterogeneity level of a new subregion to be significantly different from those observed earlier, the layer count in MARINE is increased by 1, so as to accommodate possibly a new range/group of heterogeneities. Assigning separate layers for dealing with various groups of heterogeneity explicitly resolves the catastrophic forgetting issue in MARINE. The collaborative functioning of the MARINE and the heterogeneity tackling unit is one of the key contributions of this article. It is important to note that, apart from the autonomous layer growing strategy, MARINE is also equipped with self-evolving power [1], [21] which eventually makes the model fit for online learning. The parameter learning policy, self-adaptation mechanisms, and the heterogeneity handling scheme of proposed MARINE is thoroughly described in Sections III-E1–III-E3.

1) Forward and Back-Propagation Computation for Parameter Learning: MARINE is primarily a multilayered RNN, where the hidden layers, denoted by l_k, $k = 1, \cdots L$, are arranged in ensemble fashion such that each layer is directly connected to both input and output layers. Here, L indicates the total count of layers within MARINE architecture. Initially, L is set as 1, and subsequently, the model grows/increases the layers in autonomous fashion, with an aim to handle the spatial heterogeneity as thoroughly described in Section III-E.3. The unfolded architecture of MARINE is shown in Fig. 4. Note that, at the time of learning, only one of its layers participates. The layer selection is a part of heterogeneity tackling scheme which is discussed in Section III-E.3.

In the course of processing rth subregion at the tth timestamp, $x = S(r, t)$ is fed as the input to MARINE. Consequently, $\widehat{y} = \widehat{S}(r, t+1)$ becomes the predicted output from the network, which basically corresponds to the actual output $y = S(r, t+1)$. MARINE uses rectified linear unit (ReLU) as the activation function in its hidden layers. At the end of processing, the predicted output from MARINE becomes: $\widehat{y}^{(t)} = g(O^{(t)})$, where g is the linear stretching function [10]. For any hidden layer l_k $(k = 1, \cdots L)$ of MARINE, $O^{(t)} = c + h^{(t)} V_k^\top$; $h^{(t)} = max(0, \left[b + x^{(t)} U_k^\top + y^{(t-1)} W_k^\top\right])$; U_k is the weight matrix maintaining the parameters corresponding to input-to-hidden connections, V_k is the same corresponding to hidden-to-output connections; W_k matrix maintains the parameters corresponding to output-to-hidden recurrent connections; b is the bias associated with the hidden layer, and c is the bias corresponding to the output layer.

When the predicted output is available, we estimate the loss as $\mathcal{L}(y, \widehat{y}) = (1/n) \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2$ and subsequently back-propagate this to tune the parameters/weights as follows:

$$\nabla_{V_k} \mathcal{L} = (\nabla_{O^{(t)}} \mathcal{L})^\top \cdot \left(h^{(t)}\right) \tag{6}$$

$$\nabla_{U_k} \mathcal{L} = (\nabla_{h^{(t)}} \mathcal{L})^\top \cdot x^{(t)} \tag{7}$$

$$\nabla_{W_k} \mathcal{L} = (\nabla_{h^{(t)}} \mathcal{L})^\top \cdot y^{(t-1)} \tag{8}$$

where $\nabla_{O^{i(t)}} \mathcal{L} = -q \cdot (y_i - \widehat{y}_i)$; q represents a linear stretching constant; $\nabla_{h^{i(t)}} \mathcal{L} = -q \cdot (y_i - \widehat{y}_i) \cdot V$.

2) Self-Adjustment of Within Layer Architecture: The architectural configuration of MARINE is adjusted in a self-governing manner so that it can maintain the bias-variance tradeoff [1], [21]. The bias and variance are estimated as $(E[\widehat{y}] - y)$ and $(E[\widehat{y}^2] - E[\widehat{y}]^2)$, respectively. Here, $E[\widehat{y}]$ indicates the expectation of the output, estimated as follows, considering kth layer l_k to be the participating layer:

$$E[\widehat{y}] = g(c + E[h] \cdot V_k^\top) \tag{9}$$

where

$$E[h] = b + \mu \cdot \left(U_k^\top + W_k^\top\right). \tag{10}$$

In line with the approach followed in [1] and [21], the architecture of MARINE is initialized with a single hidden layer consisting of a single hidden unit. More hidden units are automatically added one at a time in subsequent time-stamps when the model indicates an underfitting situation through its high bias. Thus, the condition for adding or growing hidden unit becomes: $\mu^t_{\text{Bias}} + \sigma^t_{\text{Bias}} \geq \mu^{\min}_{\text{Bias}} + \pi \sigma^{\min}_{\text{Bias}}$ where, μ^t_{Bias} and σ^t_{Bias} represent mean and standard deviation (s.d.) of bias at the tth time-stamp; $\mu^{\min}_{\text{Bias}}$ and $\sigma^{\min}_{\text{Bias}}$ represent those corresponding to the minimum bias. Whenever a new hidden unit is added, the associated parameters/weights are sampled from the scope of $[-1,1]$ in a random manner. In order to ensure better convergence of the model, the new parameters may also be initialized using the adaptive scope selection technique [25].

Likewise, when the model indicates an overfitting situation through its high variance, some of the hidden units are pruned from the hidden layer of the network. Thus, the condition for pruning hidden unit becomes: $\mu^t_{\text{Var}} + \sigma^t_{\text{Var}} \geq \mu^{\min}_{\text{Var}} + 2\chi \sigma^{\min}_{\text{Var}}$ where, μ^t_{Var} and σ^t_{Var} represent mean and s.d. of the variance at the tth time-stamp; $\mu^{\min}_{\text{Var}}$ and $\sigma^{\min}_{\text{Var}}$ denote the same for the minimum variance. The hidden unit, which exhibits the minimum activation so far, is ultimately selected for pruning.

To be noted, the within layer adaptation[1] is performed only in one of the layers in ensemble architecture of MARINE, which participates in the parameter learning process [1].

3) Tackling Heterogeneity Through Layer Growing in MARINE: As highlighted in Fig. 3, an explicit unit (or submodule) for tackling heterogeneity works in association with MARINE. The prime objective of this unit is to judiciously guide the recurrent network-based learning/prediction of MARINE in such a way that the catastrophic forgetting due to varying degree of spatial heterogeneity is addressed. Typically, the unit suggests MARINE to employ separate network-layers to learn the spatial relationships under different groups of heterogeneity levels, and continuously keeps track

[1]Note: While checking the unit growing/pruning condition during layer adaptation (refer Section III-E.2), both π and χ are set as $1.3\exp(-\text{bias}^2) + 0.7$, so as to attain a confidence level between 68% and 95%.

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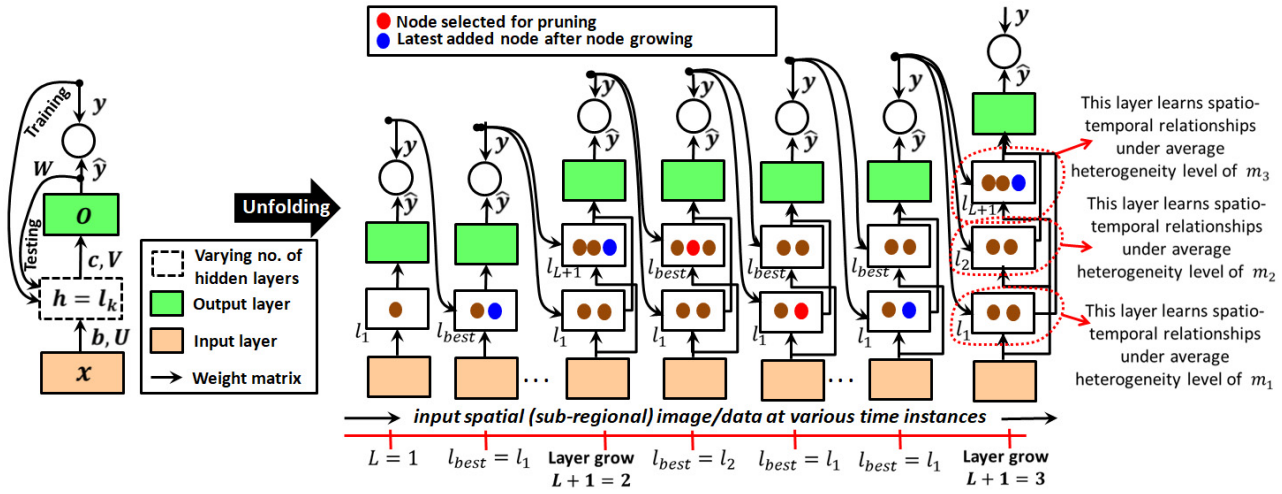


Fig. 4. MARINE architecture along with layer growing scheme for handling catastrophic forgetting due to variations in subregional heterogeneity.

of the mean/average heterogeneity level (m_k) handled by each layer ($l_k, k = 1, \ldots, L$) in MARINE. If the minimum difference between the spatial heterogeneity level a_i of a newly observed subregion s_i and the mean heterogeneity level of all the layers of MARINE is found to be larger than a threshold th, the unit instructs MARINE to create/grow another layer l_{L+1} in ensemble fashion, to deal with this new heterogeneity level. Accordingly, the mean heterogeneity level of the new layer l_{L+1} of MARINE is initialized as

$$m_{L+1} = a_i. \tag{11}$$

Otherwise, the heterogeneity tackling unit finds the best matching layer l_{best} (best $\in 1, \ldots, L$) such that $|m_{\text{best}} - a_i| < \text{th}$ is the minimum, and subsequently, updates the mean heterogeneity level m_{best} of the layer as follows:

$$m_{\text{best}} = \left(\frac{a_i + (q_{\text{best}} * m_{\text{best}})}{(q_{\text{best}} + 1)} \right). \tag{12}$$

Here, q_{best} indicates the number of observed spatial subregions which have been matched so far with the layer l_{best}, as per the similarity in associated heterogeneity levels. The l_{best} can be formally expressed as follows:

$$l_{\text{best}} = \underset{l_k, k=1,\ldots,L}{\arg\min} |m_k - a_i|, \quad |m_k - a_i| < \text{th}. \tag{13}$$

As mentioned in previous sections, the parameter learning and within-layer adaptation processes in MARINE are only performed for l_{best} (in the case of no layer growth) and for l_{L+1} (in the case layer growth occurs). Moreover, at the time of testing, the prediction is performed considering l_{best} only. Thus, irrespective of the total count of layers in MARINE, participation of only a single layer during learning/prediction and self-adaptation process makes our model lightweight in terms of computational burden. The ensemble architecture of MARINE supports realization of this scheme of single layer participation at a time.

Overall, whenever the difference between the spatial heterogeneity level of a newly observed subregion differs significantly (i.e., the difference exceeds the threshold th) from the mean heterogeneity levels associated with all the layers in MARINE, the layer-count of MARINE is increased by 1. Otherwise, the mean heterogeneity level of the best matching layer is updated according to (12). Apparently, each of the layers in MARINE indirectly helps **clustering** the spatial subregions as per their heterogeneity levels, where the mean heterogeneity levels of the layers functions as **cluster centroid**. This ensures that the previously acquired knowledge on spatial relationship is retained while accommodating the new knowledge, and thus **overcomes the catastrophic forgetting problem**. Accordingly, the proposed MARINE can be considered as an **incremental learning model** [26], [27] which can be effectively applied for spatial time series prediction over any region with almost no manual intervention.

It is important to note here that, in case the heterogeneity tackling unit finds the same layer to be the best match (l_{best}) and subsequently updates its heterogeneity level frequently, there is a high chance that the mean heterogeneity level of the layer would deviate substantially from its initial value. This may eventually allow previously found distant heterogeneity levels to be accommodated within the same layer in forthcoming instances, and hence, may again lead to the catastrophic forgetting issue. In order to combat this problem, we propose an **automated threshold updating scheme**, where, before the updating process of mean heterogeneity level as given in (12), the threshold (th) is updated as follows:

$$\text{th} = \text{th} - \left(\frac{\left(\frac{a_i + (q_{\text{best}} * m_{\text{best}})}{(q_{\text{best}} + 1)} \right) - m_{\text{best}}}{2} \right). \tag{14}$$

Thus, the heterogeneity handling in our proposed MARINE-based prediction model is achieved in **two ways**: 1) by maintaining separate hidden layers to learn spatial relationships under various groups/clusters of heterogeneity levels and 2) by availing automated threshold updating scheme that can deal with continuous variations of subregional heterogeneity over the spatial image. Note that, since the number of pixels (spatial locations) within each subregion is n, the heterogeneity level can vary within the range $[0, n]$.

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Algorithm 1 MARINE-Based-Prediction(I, T_{pred})

Input : $I = \{I_1, I_2, \cdots, I_t\}$: Series of observed images each comprising of R sub-regional data denoted by S
Input : T_{pred} = Prediction timestamp (T_{t+1})
Output: $\widehat{I}$= Predicted image comprising of predicted sub-regional data $\widehat{S}$

$L \leftarrow 1$ /* initialization of MARINE layer count */
$th \leftarrow n/2$ /* n is the pixel count in a sub-regional image */
for *each sub-regional image* S **do**
  $a \leftarrow$ Heterogeneity_level(S) /* refer Section III-C */
  if S *is the very first subregion* **then**
    $m_1 \leftarrow a$ /* initialization of mean heterogeneity level of first MARINE layer */
  end
  $l_{best} \leftarrow \arg\min_{l_k, k=1,\cdots L} |m_k - a|$
  if $|m_{best} - a| < th$ **then**
    MARINE_forward(l_{best},$S[1:t-1]$); /* refer Section III-E1 */
    MARINE_backward(l_{best},$S[1:t-1]$); /* refer Section III-E1 */
    update_threshold (th); /* refer eq. 14 */
    update_mean_heterogeneity (l_{best}); /* refer eq. 12 */
    MARINE_within_layer_adaptation(l_{best}); /* refer Section III-E2 */
  end
  else
    $l_{L+1} \leftarrow l_{best}$ /* New layer parameters are initialized with that of the best matching layer */
    MARINE_forward(l_{L+1},$S[1:t-1]$); /* refer Section III-E1 */
    MARINE_backward(l_{L+1},$S[1:t-1]$); /* refer Section III-E1 */
    $m_{L+1} \leftarrow a$
    MARINE_within_layer_adaptation(l_{L+1}); /* refer Section III-E2 */
    $L \leftarrow L + 1$ /* since a new layer is added */
  end
end
for *each sub-regional image* S **do**
  $a \leftarrow$ Heterogeneity_level(S) /* refer Section III-C*/
  $l_{best} = \arg\min_{l_k, k=1,\cdots L} |m_k - a|$
  $\widehat{S} \leftarrow$ MARINE_test(l_{best},$S[2:t]$); /* primarily this is MARINE forward propagation using data from time stamps $T_2 \cdots T_t$ */
end
return $\widehat{I}$ composed of all the predicted sub-regional image $\widehat{S}$

Accordingly, at the very beginning, we initialize the threshold value as th $= n/2$ so as to broadly categorize the subregions as highly heterogeneous or little heterogeneous, respectively. Later, with the autonomous adjustment of the threshold, the model becomes able to sophistically handle the variations in the subregional heterogeneity.

The overall process of proposed MARINE-based heterogeneity-aware prediction is presented through Algorithm 1. The additional theoretical analyses regarding the behavioral aspects of MARINE can be found in our supplementary document.

TABLE I
SUMMARY OF SPATIAL INFORMATION OF THE STUDY ZONES

Study Zones	Area	Bounding box
Zone-1 (Argentina, South America)	$\approx$ 250 km^2	Bottom-Left: 38°35'S, 65°27'W Top-Right: 36°14'S, 61°03'W
Zone-2 (Netherlands + Germany, Europe)	$\approx$ 250 km^2	Bottom-Left: 51°26'N,4°05'E Top-Right: 53°56'N, 8°02'E

IV. EXPERIMENTS

A. Dataset and Study Area

The experimentation is carried out using two series of normalized difference vegetation index (NDVI) images [28], [29], covering the South-Western part of South America, and the North-Western part of Europe, respectively, (refer Fig. 5). A higher value of NDVI indicates higher density of vegetation cover, whereas a zero or negative value of NDVI indicates no vegetation cover. Each of the NDVI time series is derived from the remote sensing images of 500 m spatial resolution, as captured by MODIS Terra satellite in the month of January for several consecutive years. These derived NDVI datasets are publicly available in the archive center of United States Geological Survey (USGS) [30] and are popularly used as benchmark in existing research works [31]. For our experimentation purpose, we have chosen two zones from the above-mentioned study areas in such a way that the heterogeneity/subregional variations of the vegetation cover in the zones are considerably different. A summary of the selected study zones, here after termed as "Zone-1" and "Zone-2," are summarized in Table I. The percentage distributions of various degrees of heterogeneity in Zone-1 and Zone-2, as calculated in terms of range-value [3], can be visualized from Fig. 6. Note that, a significant percentage (30%) of the subregions within Zone-2 has the lowest range value of 0.5 km, whereas the minimum range value of the subregions within Zone-1 is 1 km. Consequently, Zone-2 can be considered to be far more heterogeneous than Zone-1. We do not strictly use land-cover product for measuring the heterogeneity, because ours is a generic model which aims at dealing with heterogeneity introduced by the working variable, which may be NDVI, LST, soil moisture, etc., associated with the spatial image time series. In our case study, the working variable is NDVI that implicitly signifies various land-cover types, including vegetation, water-body, built-up area, and so on. The annual NDVI time series over Zone-1 is collected for the duration of 2011–2018 and that over Zone-2 is collected for the duration of 2013–2020. Note that, the MODIS Terra satellite images are practically available on every 16-day basis.

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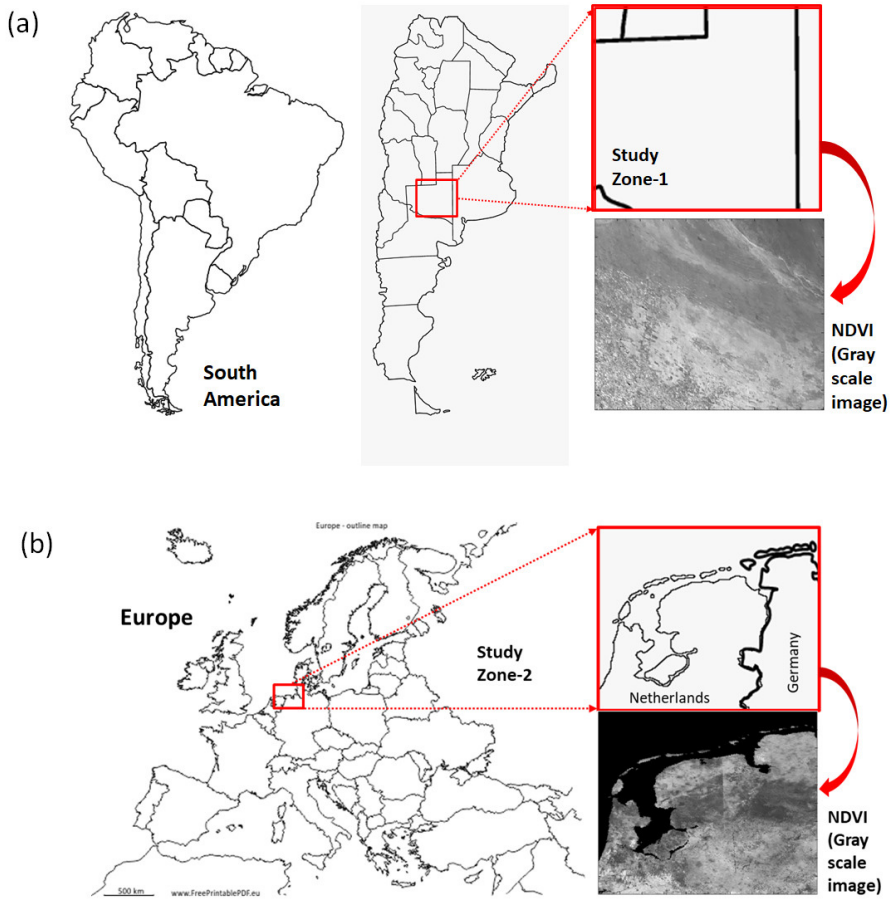


Fig. 5. Study zones considered for the case study: (a) study Zone-1 (Argentina, South America) and (b) study Zone-2 (Netherlands+Germany, Europe).

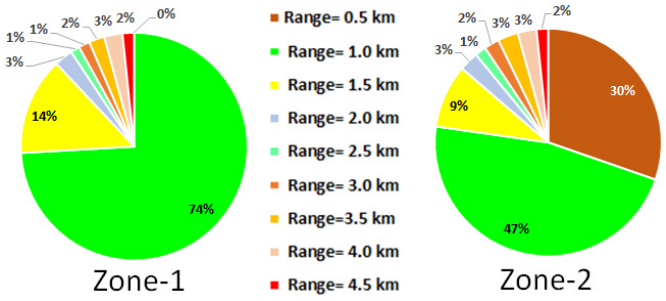


Fig. 6. Percentage distribution of subregional variance/heterogeneity at various degree/level (denoted by the range value).

However, we experiment only with the annual data, since we aim to predict changes other than seasonal effect.

B. Baselines and Experimental Settings

Our proposed MARINE is evaluated in comparison with *seven* state-of-the-art deep NN models as briefly discussed below.

1) *DNN:* [32]: This deep neural network model is primarily influenced from the work of Aparna *et al.* [32]. In order to form a deeper architecture, we have considered three fully connected layers in the network architecture.
2) *Deep-STEP:* [10]: This is specifically designed to achieve spatio-temporal prediction of remote sensing image time series. The model uses a deep stacking architecture. In order to handle the heterogeneity issue, we have applied individual local version of this model for each subregion.
3) *CNN:* [16]: This model of CNN is influenced from the work of Han *et al.* [16]. In this model, though the convolution operation is applied on local field of view, the model finally tries to perceive the global view using fully connected network.
4) *CNN-LSTM$_1$:* This model is a combination of CNN and LSTM, as used by Sun *et al.* [17] and designed with the help of Conv2D, LSTM, and Fully-connected/Dense layers from Keras. Similar to the work of Yang *et al.* [11] we have applied individual local version of this model on subregional basis to deal with the heterogeneity issue.
5) *CNN-LSTM$_2$:* This is another model of combined CNN and LSTM as employed in the work of Nevavuori *et al.* [18] and designed with the help of Conv2D, LSTM, and Fully-connected/Dense layers from Keras.
6) *LSTM-MLP:* [15]: This model is similar to the one used by Jahanbakht *et al.* [15]. The deep network architecture here is comprised of stacked LSTM-MLP networks.

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TABLE II

OVERVIEW OF NETWORK CONFIGURATION AND OTHER HYPERPARAMETERS OF THE CONSIDERED MODELS. [t: #OBSERVED TIMESTAMPS; n: #PIXEL IN INPUT IMAGE/IMAGE PATCH; sf AND sh: SEQUENCE NUMBERS OF FULLY-CONNECTED AND HIDDEN LAYER; SGD: STOCHASTIC GRADIENT DESCENT]

Model	Hyper-parameters				
	Hidden-layer configuration		Learning Rate	Optimizer	Epochs
	Layer count	Unit count per layer			
NARNET	Fully-Connected (F): 2	10	0.001	LMA	200
DNN	Fully-Connected (F): 3	F: $\{(n\text{–}1) \times 2\} \times 2^{sf}$	0.001	SGD	200
Deep-STEP	Deep stacking layer [10]: t ;	n	0.5	SGD	200
CNN	Convolutional (CL): 5; Fully-Connected (F): 4	CL: $\{(n\text{–}1) \times 2\} \times 2^{sh}$; F: $\{(n\text{–}1) \times 2\} \times 2^{sf}$	0.001	Adam	200
CNN-LSTM$_1$	Convolutional (CL): 2; LSTM: 1; Fully-Connected (F): 1	CL: $\{(n\text{–}1) \times 2\} \times 2^{sh}$; LSTM: $(n\text{–}1) \times 2$ F: $(n\text{–}1) \times 2$	0.5	Adam	200
CNN-LSTM$_2$	Convolutional (CL): 2; LSTM: 2; Fully-Connected (F): 1	CL: $\{(n\text{–}1) \times 2\} \times 2^{sh}$; LSTM: $(n\text{–}1) \times 2$ F: $(n\text{–}1) \times 2$	0.5	Adadelta, Adam	200
LSTM-MLP	LSTM: 2; Fully-Connected (F): 2	LSTM: $(n\text{–}1) \times 2$; F: $\{(n\text{–}1) \times 2\} \times 2^{sf}$	0.100	SGD	200
SARDINE	Self/Auto-decided; Begins with a single layer	Self/Auto-Adjusted; Begins with a single layer	0.001	SGD	1 epoch per sub-regional data
Proposed MARINE	Self/Auto- decided; Begins with a single layer	Self/Auto- Adjusted; Begins with a single layer	0.001	SGD	1 epoch per sub-regional data

7) *SARDINE:* [1]: This is a recently proposed model based on autonomous learning scheme. Comparative study with SARDINE is necessary, since, likewise our presently proposed model, SARDINE is also self-adaptive. However, SARDINE possesses hierarchically deep architecture.

In addition to these state-of-the-art deep network models, MARINE is also compared with two traditional benchmark models, namely, the spatio-temporal ordinary Kriging (ST-OK) [8], and the nonlinear auto-regressive neural network model (NARNET) [33]. ST-OK is a conventional statistical model and can be considered as the spatio-temporal extension of ordinary Kriging. It is extensively used for spatio-temporal prediction of missing values in satellite remote sensing imagery. On the other side, NARNET is a variant of nonlinear auto-regressive neural network model which takes into account spatial features in the form of exogenous variables and originally belongs to nondeep family. NARNET is also widely used for time series prediction. In our experiment, the NARNET model has been trained using Levenberg–Marquardt algorithm (LMA) in MATLAB R2020a environment, whereas for the ST-OK we have used the in-built krigeST model from the gstat package of the R Tool.[2] All the models, including our proposed MARINE, are executed in Windows 64-bit operating system (CPU processor: 3.4 GHz; RAM: 64 GB). For the NDVI dataset over Zone-1, the training is conducted using the image time series of the duration 2011–2017 and the prediction is performed for the year 2018, whereas for the NDVI dataset over Zone-2, the training is conducted using the image time series of the duration 2013–2019 and the prediction is performed for the year 2020. In both the cases, the area of each subregion is considered to be 1.5 km^2 (approx.). For each baseline, expect SARDINE, the network structure (i.e., the proportions of convolutional layers, LSTM layers, and fully-connected layers) is fixed as per the configuration mentioned in the respective source code or article. However, if the model prediction is found to be unexpectedly poor, we empirically adjust the associated hyperparameters (hidden layer count, hidden unit count, number of epochs, learning rate, etc.) of these baselines, to account for their best performances. It is interesting to note here that, for both SARDINE and the proposed MARINE-based prediction, the hidden layer configuration is auto-adjusted in self-governing fashion, with no requirement for empirical analysis. The network structure and the other hyperparameters for all the considered models are summarized in Table II.

[2]https://cran.r-project.org/web/packages/gstat/index.html

C. Performance Metrics

The performance assessments for the models are carried out using three commonly used goodness-of-fit-criteria: normalized root mean square deviation (NRMSD), mean absolute error (MAE), and peak signal-to-noise ratio (PSNR) [34]. Mathematically, these can be presented as follows:

$$\text{NRMSD} = \frac{1}{(O_{\max} - O_{\min})} \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\mathcal{V}_{o_i} - \mathcal{V}_{p_i})^2} \quad (15)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |\mathcal{V}_{o_i} - \mathcal{V}_{p_i}| \quad (16)$$

$$\text{PSNR} = 20.\log_{10}\left(\frac{\text{MAX}_I}{\sqrt{\text{MSE}}}\right) \quad (17)$$

where N denotes the total pixel count in a spatial image, $O_{\max}$ denotes the maximum observed value of pixel corresponding to the image, $O_{\min}$ is the minimum value observed for the same, $\mathcal{V}_{o_i}$ denotes the actual/observed value at the ith pixel position of the image, $\mathcal{V}_{p_i}$ is the corresponding predicted value, MAX_I represents the maximum possible value of any pixel in the image, and MSE indicates the mean squared error.

NRMSD and MAE are used as the measures of prediction errors. Ideally, in case the predicted image perfectly matches

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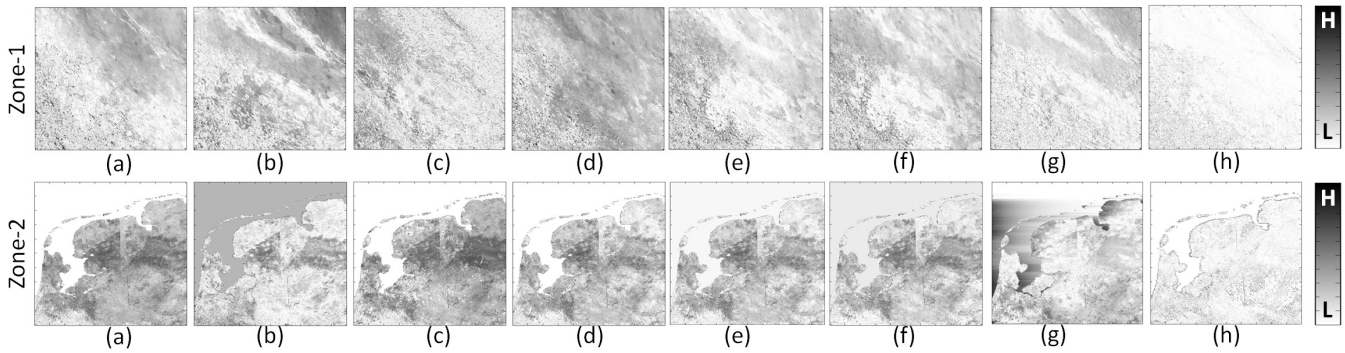


Fig. 7. Normalized error surfaces corresponding to the images predicted by (a) DNN, (b) Deep-STEP, (c) CNN, (d) CNN-LSTM$_1$, (e) CNN-LSTM$_2$, (f) LSTM-MLP, (g) SARDINE, and (h) MARINE (proposed).

TABLE III
COMPARATIVE STUDY OF PREDICTION PERFORMANCE OF PROPOSED MARINE [BOLDFACE INDICATES THE BEST VALUE]

Study Zones	Metrics	ST-OK	NARNET	DNN	Deep-STEP	CNN	CNN-LSTM$_1$	CNN-LSTM$_2$	LSTM-MLP	SARDINE	MARINE (Proposed)
Zone-1	MAE	0.09	0.11	0.18	0.11	0.19	0.19	0.10	0.07	0.05	**0.04**
	NRMSD	0.18	0.25	0.11	0.11	0.12	0.10	0.06	0.05	0.04	**0.03**
	PSNR	14.9	11.9	19.1	19.2	18.6	19.9	23.9	26.6	28.6	**29.6**
Zone-2	MAE	0.26	0.16	0.41	0.26	0.24	0.19	0.19	0.17	0.12	**0.06**
	NRMSD	0.25	0.21	0.27	0.09	0.19	0.13	0.13	0.11	0.08	**0.05**
	PSNR	11.9	13.5	11.2	20.5	14.4	17.6	17.8	18.8	21.6	**24.9**

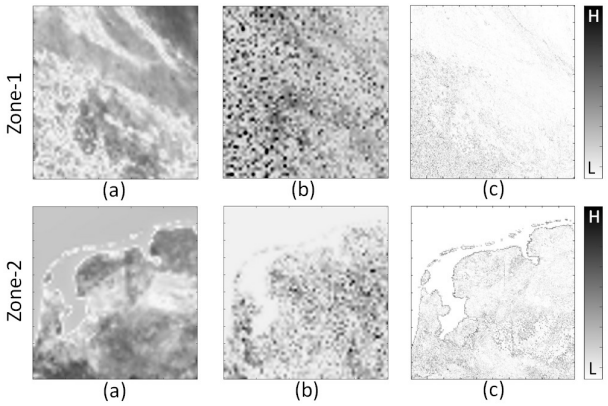


Fig. 8. Normalized error surfaces corresponding to the images predicted by (a) ST-OK, (b) NARNET, (c) MARINE (proposed).

with the observed one, the values of NRMSD and MAE become 0. PSNR value (in decibel) further indicates the predicted image quality. The higher the value of PSNR, the better the quality of the predicted image. In addition, the proposed MARINE is also evaluated with regard to execution time requirement (in seconds) for further analyzing the prediction overhead due to its heterogeneity handling capability.

D. Results and Discussions

The results of experimentation are summarized in Tables III and IV and in Figs. 7–12. On analyzing the outcomes, we can draw inference on the following bases.

1) Prediction Accuracy: In our experimentation, the accuracy is measured in terms of error or deviation with respect to the actual value for each pixel of the predicted image. It may be noted from Table III that the error measures (NRMSD and MAE) corresponding to our presently proposed MARINE are notably less compared to other baselines, for both the study zones. More interestingly, even with the increased level of spatial heterogeneity in Zone-2, the prediction error corresponding to proposed MARINE remains little altered, in contrast to that of the others. This evidently demonstrates the effectiveness of the heterogeneity-aware scheme in MARINE. The superiority of the proposed MARINE can also be visualized from the normalized error surfaces corresponding to the respective predicted images as depicted in Figs. 7 and 8. Incidentally, though the conventional ST-OK and NARNET models are found to show nearly comparable performances to that of DNN (with three fully-connected layers), these are achieved with the expense of substantial computational time as summarized in Table IV.

2) Predicted Image Quality: The quality of the predicted images, as measured in terms of PSNR, is summarized in Table III. It may be observed that the PSNR values corresponding to the proposed MARINE range in between $\approx$25 and $\approx$30 dB, indicating quite a promising quality of the predicted images. The minimum percentage improvement of the prediction quality of the proposed model is 3% over the state-of-the-art techniques considered. More importantly, the percentage improvement raises to a minimum value of 15% in case the spatial zone (Zone-2) has higher degree of heterogeneity. In other words, the prediction performance of MARINE is less affected by increasing levels of spatial heterogeneity, compared to the others. Note that, among the various baselines, the performances of NARNET, Deep-STEP, and CNN-LSTM$_1$ are less deviated at the time of prediction for the Zone-2, compared to Zone-1. This is so, because these

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TABLE IV

COMPARATIVE STUDY OF EXECUTION TIME OF MARINE (PROPOSED) AND THE CONSIDERED BASELINES

Study Zones	Execution time (in sec.) for the considered prediction models									
	ST-OK	NARNET	DNN	Deep-STEP	CNN	CNN-LSTM$_1$	CNN-LSTM$_2$	LSTM-MLP	SARDINE	MARINE (Proposed)
Zone-1	6013	1422×10^2	203.309	683.802	9927.149	1876.618	1828.05	1583.797	969.507	533.262
Zone-2	5429	1389×10^2	114.561	660.488	9917.839	1737.343	1671.22	1520.561	965.623	516.002

For all the baselines, except SARDINE, the time for empirical analysis to find out the best fit architecture is not considered. However, note that, being based on autonomous learning scheme, the structure adjustment times for both SARDINE and MARINE are included by default in the execution time.

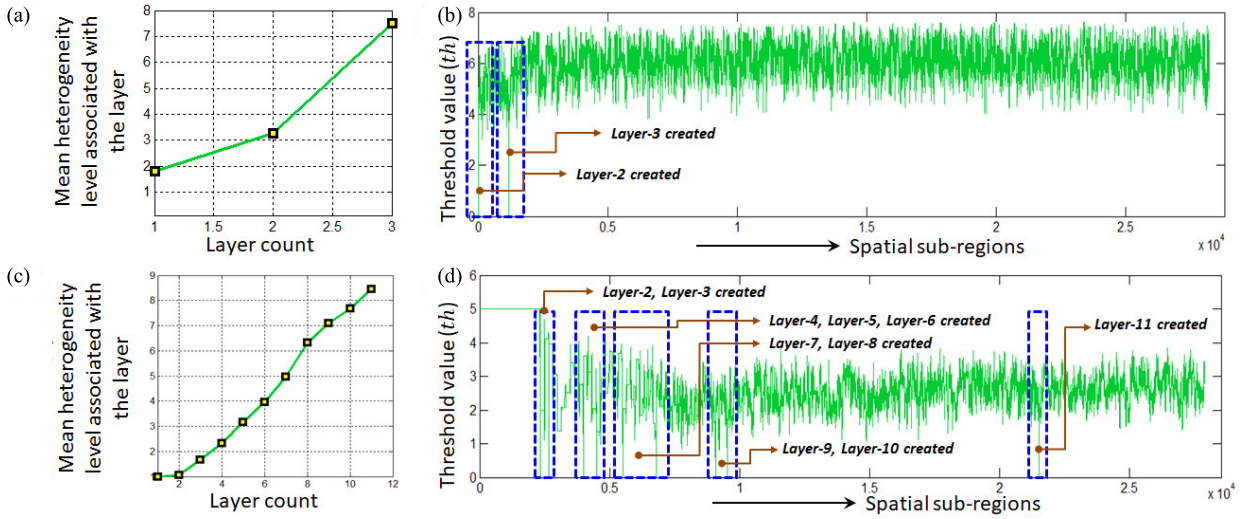


Fig. 9. Autonomous layer growth and threshold (th) adjustment in MARINE in the course of prediction for Zone-1 [(a) and (b)] and Zone-2 [(c) and (d)].

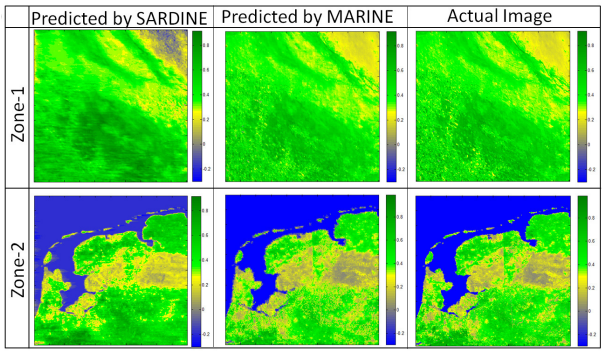


Fig. 10. Actual image versus predicted image by SARDINE and MARINE.

three models have been employed along with the individual local versions for each subregion. Nevertheless, our proposed MARINE substantially outperforms all these models, in terms of overall prediction performance.

3) Heterogeneity Handling Ability: The substantially improved prediction performance of MARINE, as discussed-above, demonstrates the effectiveness of its heterogeneity-aware prediction scheme. The auto-construction of multilayered ensemble architecture to separately handle varying degrees of spatial heterogeneity helps MARINE to actively combat the catastrophic forgetting issue, thus leading to promising prediction performance for both the study zones. The typical view of autonomous layer growth and threshold (th) adjustment in MARINE is depicted in Fig. 9. Each time a growth/addition of layer is driven by the automatic fall of threshold values which may be visualized within the dotted blue boxes indicated in the figure. It is interesting to note here that the higher the variation of the heterogeneity levels within a study zone, the more the tendency of MARINE in growing the layers to avoid the catastrophic forgetting of acquired knowledge regarding previously observed spatial relationships. However, as proved in Lemma-4 (refer supplementary document), the maximum number of layers in the network can never go beyond $\lceil R/2 \rceil$, where R is the number of subregions within a study zone.

4) Computational Overhead: It may be noticed from the methodology section that the overhead of our proposed model primarily remains in its heterogeneity handling scheme, which requires additional computation for measuring spatial/spatio-temporal variation, and also, in additional storage requirement for storing the parameters associated with the growing layers. However, the single pass as well as autonomous learning capability of MARINE helps the model in substantially compensate for the overall time requirement (refer Table IV). The autonomous growth of layers, auto-adjustment of threshold values, and self-construction of within layer architecture of MARINE can be visualized from Figs. 9 and 12. All these autonomous learning traits eliminate the requirement for any kind of empirical study to prefix the architectural configuration of MARINE, thus leading to save of significant amount of time. Furthermore, since irrespective of the count of layer

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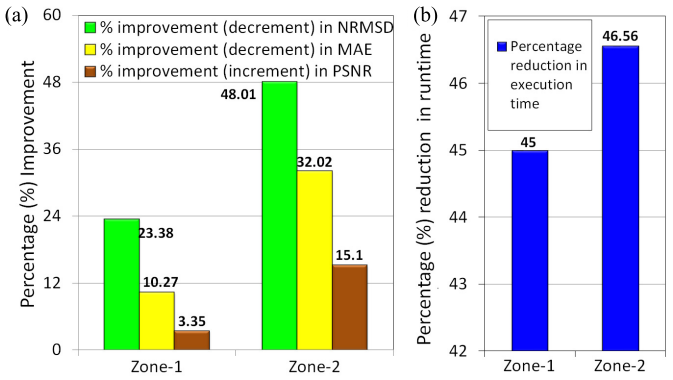


Fig. 11. SARDINE versus MARINE: Comparative performance regarding (a) prediction quality and (b) runtime.

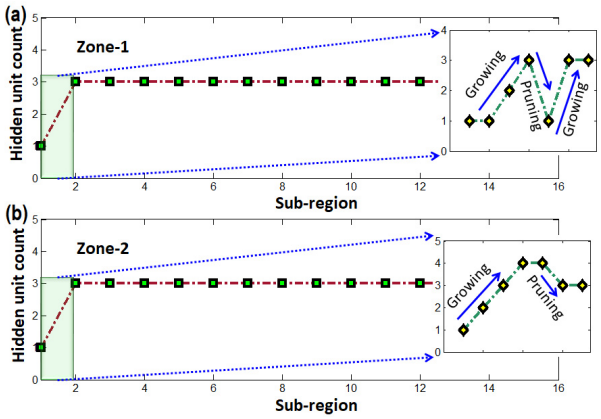


Fig. 12. Typical scenario of hidden unit growing and pruning within a particular network layer in MARINE for (a) Zone-1 and (b) Zone-2.

in MARINE architecture, each time only one of its layers take part in training/testing, and since MARINE requires no explicit computation for feature set preparation, the model succeeds in notably outperforming the autonomous state-of-the-art, SARDINE, as well (see Table IV, Figs. 10 and 11).

Overall, the proposed MARINE with its unique ability of autonomously handling heterogeneity is found to outperform all the considered baselines. The minimum percentage improvement of MARINE over the best-performing baseline SARDINE is 22%, 10%, and 3%, in NRMSD, MAE, and PSNR, respectively (refer Figs. 10 and 11). The performance improvement is more strongly reflected when the heterogeneity level of the study zone increases. Furthermore, as discussed in Section IV-D.4, the single layer participation scheme and the nonrequirement of feature set preparation leads to ≈45% reduction in computational time for MARINE over the SARDINE model.

E. Ablation Study

It may be noted that all the four major components (i.e., subregional division of data, measuring spatial/spatio-temporal heterogeneity, vector representation of the subregional data, and finally, heterogeneity-aware learning and prediction using MARINE) in our proposed model are integrated parts of the framework. Consequently, the final task of learning/prediction cannot be accomplished if any combination of these modules is removed. The only option for doing ablation study is to remove together the heterogeneity management module and the heterogeneity tackling submodule from the proposed framework. In that case, the learning mechanism as well as the accuracy of the model would be no better than SARDINE [1], the comparative study with which is already included in Section IV-D.4 (refer Table IV, Figs. 10–11).

V. Conclusions and Future Work

In this article, we have introduced a novel framework for spatio-temporal prediction with special focus toward autonomously handling catastrophic forgetting issue that arises due to the presence of varying degrees of spatial heterogeneity. The key contribution of this framework is MARINE, which combats the issue of catastrophic forgetting by autonomously generating ensemble of recurrent learning layers to separately deal with diverse levels of heterogeneity. Empirically, we show that our heterogeneity-aware ensemble learning scheme considerably improves the prediction performance compared to the state-of-the-art deep network models.

It is interesting to note that, though we have evaluated our proposed model using NDVI image time series, our model can be directly employed for spatio-temporal prediction of the series of derived remote sensing images over other variables, such as normalized difference built-up index (NDBI), green chlorophyll index (GCI), and LST, as well, with no further modification. Accordingly, our proposed model can be used in several interesting applications, including urban sprawl detection, crop field monitoring, land-cover change analysis, and weather prediction. Furthermore, ample scope remains in future to parallelizing the computation of subregional variances, and thereby, making the model more suitable for (near) real-time prediction.

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